P4.18

let $\begin{cases} L_+ f_\ell^m = A_\ell^m f_\ell^{m+1} \\ L_- f_\ell^m = B_\ell^m f_\ell^{m-1} \end{cases}$ find A_ℓ^m, B_ℓ^m.

Use the fact that $\begin{cases} L^2 f_\ell^m = \hbar \ell(\ell+1) f_\ell^m \\ L_z f_\ell^m = \hbar m f_\ell^m \end{cases}$

and $L^2 = L_\pm L_\mp + L_z^2 \mp \hbar L_z$

To find A_ℓ^m form: $L^2 f_\ell^m = (L_- L_+ + L_z^2 + \hbar L_z) f_\ell^m$

$$\Rightarrow \langle f_\ell^m | L^2 f_\ell^m \rangle = \langle f_\ell^m | L_- L_+ f_\ell^m \rangle +$$

$$+ \langle f_\ell^m | L_z^2 f_\ell^m \rangle + \hbar \langle f_\ell^m | L_z f_\ell^m \rangle =$$

$$= \langle L_-^\dagger f_\ell^m | L_+ f_\ell^m \rangle + (\hbar m)^2 \langle f_\ell^m | f_\ell^m \rangle + \hbar \cdot \hbar m \langle f_\ell^m | f_\ell^m \rangle$$

But $L_-^\dagger = (L_x - iL_y)^\dagger = L_x + iL_y = L_+$ because $L_x^\dagger = L_x$, $L_y^\dagger = L_y$

$$\Rightarrow \hbar^2 \ell(\ell+1) = \underbrace{\langle L_+ f_\ell^m | L_+ f_\ell^m \rangle}_{\langle A_\ell^m f_\ell^{m+1} | A_\ell^m f_\ell^{m+1} \rangle = |A_\ell^m|^2} + \hbar^2 m^2 + \hbar^2 m$$

$$\Rightarrow |A_\ell^m|^2 = \hbar^2 (\ell(\ell+1) - m(m+1))$$

$$A_\ell^m = \hbar \sqrt{\ell(\ell+1) - m(m+1)}$$

Thus for $m = \ell$: $A_\ell^\ell = \hbar \sqrt{\ell(\ell+1) - \ell(\ell+1)} = 0$ so that $L_+ f_\ell^\ell = 0$ as it should be.

Similarly, taking $L^2 = L_+ L_- + L_z^2 - \hbar L_z$ we find that

$$|B_\ell^m|^2 = \hbar^2 (\ell(\ell+1) - m(m-1)) \; ; \; B_\ell^m = \hbar \sqrt{\ell(\ell+1) - m(m-1)}$$

so that for $m = -\ell$: $L_- f_\ell^{-\ell} = \hbar \sqrt{\ell(\ell+1) - (-\ell)(-\ell-1)} f_\ell^{-\ell-1} = 0.$

P4.22 (a) $L_+ Y_\ell^\ell = \hbar\sqrt{\ell(\ell+1) - \ell(\ell+1)}\; Y_\ell^{\ell+1} = 0.$

(b) Eq. 4.130 : $L_+ = \hbar e^{+i\varphi}\left(\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\varphi}\right)$

$$L_z = \frac{\hbar}{i}\frac{\partial}{\partial\varphi}$$

Since $L_z Y_\ell^\ell(\theta,\varphi) = \hbar\ell Y_\ell^\ell(\theta,\varphi)$

$$\Rightarrow \frac{\hbar}{i}\frac{\partial}{\partial\varphi}Y_\ell^\ell(\theta,\varphi) = \hbar\ell Y_\ell^\ell(\theta,\varphi)$$

$$\Rightarrow Y_\ell^\ell(\theta,\varphi) = F(\theta)\, e^{i\ell\varphi}$$

Then $L_+ Y_\ell^\ell(\theta,\varphi) = \hbar e^{i\varphi}\left(\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\varphi}\right) F(\theta)\, e^{i\ell\varphi} = 0$

$$\Rightarrow e^{i\varphi} e^{i\ell\varphi}\left[\frac{\partial}{\partial\theta} + i\cot\theta\,(i\ell)\right]F(\theta) = 0$$

$$\Rightarrow \frac{dF}{d\theta} = \ell\cot\theta\, F(\theta) = \ell\frac{\cos\theta}{\sin\theta}F(\theta)$$

$$\Rightarrow \frac{dF}{F} = \ell\frac{\cos\theta}{\sin\theta}d\theta = \ell\frac{d(\sin\theta)}{\sin\theta} = d\left(\ln(\sin\theta)\right)\cdot\ell$$

$$\Rightarrow \ln F(\theta) = \ell\ln(\sin\theta) = \ln(\sin\theta)^\ell$$

$$\Rightarrow F(\theta) = (\sin\theta)^\ell \cdot C_\ell$$

normalization constant

Thus $\boxed{Y_\ell^\ell(\theta,\varphi) = C_\ell\,(\sin\theta)^\ell\, e^{i\ell\varphi}}$

(c) Require $1 = \int |Y_\ell^\ell|^2 \sin\theta\, d\theta\, d\varphi \Rightarrow$

$$\Rightarrow 1 = |C_\ell|^2 \int_0^\pi (\sin\theta)^{2\ell}\sin\theta\, d\ell \int_0^{2\pi} d\varphi = 2\pi|C_\ell|^2 \cdot$$

$$\int_{-1}^{1}(\sin^2\theta)^\ell\, d(-\cos\theta)$$

Introduce $x = -\cos\theta$, $dx = d(-\cos\theta) = \sin\theta\, d\theta$

$$\sin^2\theta = 1 - \cos^2\theta = 1 - x^2$$

$$\Rightarrow \int_{-1}^{1} (\sin^2\theta)^\ell \, d(-\cos\theta) = \int_{-1}^{1} (1-x^2)^\ell \, dx$$

$$\Rightarrow 1 = 2\pi |C_\ell|^2 \cdot \underbrace{2\int_0^1 (1-x^2)^\ell \, dx}_{I(\ell)}$$

let $y = x^2 \Rightarrow dy = 2x\,dx \Rightarrow dx = \dfrac{dy}{2\sqrt{y}}$

$$\Rightarrow I(\ell) = 2\cdot\int_0^1 (1-x^2)^\ell \, dx = \int_0^1 (1-y)^\ell \, y^{-1/2}\, dy =$$

$$= \int_0^1 (1-y)^{\ell+1-1} \, y^{1/2-1} \, dy = B\left(\ell+1, \tfrac{1}{2}\right)$$

where beta-function $B(a,b) \equiv \dfrac{\Gamma(a)\,\Gamma(b)}{\Gamma(a+b)}$ and Γ = gamma function

Thus $I(\ell) = \dfrac{\Gamma(\ell+1)\,\Gamma(\frac{1}{2})}{\Gamma(\ell+1+\frac{1}{2})} = \dfrac{\Gamma(\frac{1}{2})\,\Gamma(1+\ell)}{\Gamma(\frac{3}{2}+\ell)}$

Now, $\Gamma(n) = (n-1)!$; $\cancel{\Gamma(\tfrac{1}{2}+n) = \dfrac{\Gamma(2n)}{\Gamma(n)}\dfrac{\sqrt{2\pi}}{2^{2n-1/2}}}$, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

$\Gamma(\frac{1}{2}+n) = \sqrt{\pi}\,\dfrac{(2n-1)!!}{2^n}$

$$\Rightarrow I(\ell) = \frac{\sqrt{\pi}\;\ell!}{\sqrt{\pi}\;\dfrac{(2(1+\ell)-1)!!}{2^{1+\ell}}} = \frac{2^{1+\ell}\,\ell!}{(2\ell+1)!!}$$

$\Gamma(\frac{3}{2}+\ell) = \Gamma(\frac{1}{2}+1+\ell)$

But double factorial $(2\ell+1)!! = \dfrac{(2\ell+1)!}{2^\ell \ell!}$

$$\Rightarrow I(\ell) = \frac{2^{2\ell+1}(\ell!)^2}{(2\ell+1)!} \Rightarrow C_\ell = \frac{1}{\sqrt{2\pi}}\sqrt{\frac{(2\ell+1)!}{2^{2\ell+1}(\ell!)^2}}$$ agrees with Table 4.3

$\ell=0 \Rightarrow C_\ell = \left(\dfrac{1}{4\pi}\right)^{1/2}$; $\ell=1 \Rightarrow C_\ell = \left(\dfrac{3!}{4\pi\cdot 2^2}\right)^{1/2} = \left(\dfrac{3}{8\pi}\right)^{1/2}$; $\ell=2 \Rightarrow C_\ell = \sqrt{\dfrac{5!}{4\pi\cdot 2^4 (2!)^2}} = \sqrt{\dfrac{15}{32\pi}}$; $\ell=3 \Rightarrow C_\ell = \sqrt{\dfrac{7!}{4\pi\cdot 2^6 (3!)^2}}$ $\boxed{35}$

P4.21 $L_\pm = \pm\hbar e^{\pm i\varphi}\left(\frac{\partial}{\partial\theta} \pm i\cot\theta\frac{\partial}{\partial\varphi}\right)$

(a) Derive eq. 4.131 by acting with $\hat{L}_+\hat{L}_-$ on arbitrary $f(\theta,\varphi)$:

$$\hat{L}_+\hat{L}_- f = -\hbar^2 e^{i\varphi}\left(\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\varphi}\right) e^{-i\varphi}\left(\frac{\partial}{\partial\theta} - i\cot\theta\frac{\partial}{\partial\varphi}\right) f$$

$$= -\hbar^2 e^{i\varphi}\left(\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\varphi}\right) e^{-i\varphi}\left(\frac{\partial f}{\partial\theta} - i\cot\theta\frac{\partial f}{\partial\varphi}\right) =$$

$$= -\hbar^2 e^{i\varphi}\left\{ e^{-i\varphi}\frac{\partial^2 f}{\partial\theta^2} - ie^{-i\varphi}\frac{\partial}{\partial\theta}\left(\cot\theta\frac{\partial f}{\partial\varphi}\right) + i\cot\theta \cdot \right.$$

$$\left. \cdot \frac{\partial}{\partial\varphi}\left(e^{-i\varphi}\frac{\partial f}{\partial\theta}\right) + \cot^2\theta\frac{\partial}{\partial\varphi}\left(e^{-i\varphi}\frac{\partial f}{\partial\varphi}\right)\right\} =$$

$$= -\hbar^2 e^{i\varphi}\left\{ e^{-i\varphi}\frac{\partial^2 f}{\partial\theta^2} - ie^{-i\varphi}\left(\frac{-1}{\sin^2\theta}\frac{\partial f}{\partial\varphi} + \cot\theta\frac{\partial^2 f}{\partial\theta\partial\varphi}\right)\right.$$

$$+ i\cot\theta\left(-ie^{-i\varphi}\frac{\partial f}{\partial\theta} + e^{-i\varphi}\frac{\partial^2 f}{\partial\varphi\partial\theta}\right) + \cot^2\theta \cdot$$

$$\left. \cdot\left(-ie^{-i\varphi}\frac{\partial f}{\partial\varphi} + e^{-i\varphi}\frac{\partial^2 f}{\partial\varphi^2}\right)\right\} = -\hbar^2 e^{i\varphi} \cdot$$

$$\cdot\left\{ e^{-i\varphi}\frac{\partial^2 f}{\partial\theta^2} + ie^{-i\varphi}\left(\frac{1}{\sin^2\theta} - \cot^2\theta\right)\frac{\partial f}{\partial\varphi} + e^{-i\varphi}\cot\theta\frac{\partial f}{\partial\theta}\right.$$

$$\left. + e^{-i\varphi}\cot^2\theta\frac{\partial^2 f}{\partial\varphi^2} + ie^{-i\varphi}\left(-\cot\theta + \cancel{\cot\theta}\right)\frac{\partial^2 f}{\partial\varphi\partial\theta}\right\}$$

$$= -\hbar^2\left\{\frac{\partial^2 f}{\partial\theta^2} + \cot\theta\frac{\partial f}{\partial\theta} + \cot^2\theta\frac{\partial^2 f}{\partial\varphi^2} + i\underbrace{\left(\frac{1}{\sin^2\theta} - \frac{\cos^2\theta}{\sin^2\theta}\right)}_{1}\frac{\partial f}{\partial\varphi}\right\}$$

Thus, $\hat{L}_+\hat{L}_- = -\hbar^2\left\{\frac{\partial^2}{\partial\theta^2} + \cot\theta\frac{\partial}{\partial\theta} + \cot^2\theta\frac{\partial^2}{\partial\varphi^2} + i\frac{\partial}{\partial\varphi}\right\}$

(b) Derive eq. 4.132 : from eq. 4.112

$$\hat{L}^2 = \hat{L}_+ \hat{L}_- + \hat{L}_z^2 - \hbar \hat{L}_z$$

$$\Rightarrow \hat{L}^2 = -\hbar^2 \left\{ \frac{\partial^2}{\partial\theta^2} + \cot\theta \frac{\partial}{\partial\theta} + \cot^2\theta \frac{\partial^2}{\partial\varphi^2} + i\frac{\partial}{\partial\varphi} \right\}$$

$$+ \left(\frac{\hbar}{i}\frac{\partial}{\partial\varphi}\right)^2 - \hbar \frac{\hbar}{i}\frac{\partial}{\partial\varphi} = -\hbar^2 \left\{ \frac{\partial^2}{\partial\theta^2} + \cot\theta \frac{\partial}{\partial\theta} \right.$$

$$\left. + \underbrace{(\cot^2\theta + 1)}_{\frac{1}{\sin^2\theta}} \frac{\partial^2}{\partial\varphi} + \cancel{i\frac{\partial}{\partial\varphi} - i\frac{\partial}{\partial\varphi}} \right\} =$$

$$= -\hbar^2 \left\{ \frac{\partial^2}{\partial\theta^2} + \cot\theta \frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2} \right\}.$$

Observe now that $\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta \frac{\partial}{\partial\theta}\right) =$

$$= \frac{1}{\sin\theta}\left(\cos\theta \frac{\partial}{\partial\theta} + \sin\theta \frac{\partial^2}{\partial\theta^2}\right) = \cot\theta \frac{\partial}{\partial\theta} + \frac{\partial^2}{\partial\theta^2}$$

Thus $\hat{L}^2$ simplifies to

$$\hat{L}^2 = -\hbar^2 \left\{ \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta \frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2} \right\}.$$